CES-FL: proposed upgrade

Partial fine-tuning + group-robust classification + controlled age invariance

PROPOSED – PERFORMANCE NOT YET VALIDATED

Inference architecture (no demographic metadata at inference)

A. Input image



224 × 224 × 3
(illustrative only)

B. Detection-pretrained encoder

Early layers: frozen (pretrained)

Final stage: trainable (partial fine-tuning)



Xception / ResNet-50 • BatchNorm frozen

Spatial features F



$H \times W \times C$

C1. Global branch

$g = \text{mean}(F)$ → Global MLP → z_g; Sigmoid gate → r_g; (z_g, r_g)

C2. Local branch



Patches from F ($HW \times C$)

Shared 1 × 1 patch MLP → z_p; Sigmoid patch gate → r_p; (z_p, r_p)

D. Positive gated fusion

$$M = a_p r_p \odot z_p$$

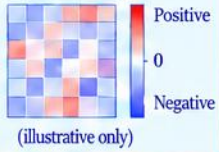
$$z = a_g r_g z_g + \text{mean}(M) + b$$

$$a_g, a_p > 0$$

E. Outputs

Real / fake score

Signed LOCAL contribution M



(illustrative only)

Gates are learned weights; M is not a forgery mask.

TRAIN-only upgrades — evaluate as separate ablations

1. Group-robust classification

Original TRAIN images + labels + group metadata



Group DRO on class × demographic groups

Regularization + validation selection

Demographic groups (e.g., age, gender, ethnicity)

Metadata used only in this training module (not fed to the encoder)

2. Controlled age pairs

Original → independent older / younger edits



Original (identity i)

Edit to older → Older edit (same identity i)



Edit to younger → Younger edit (same identity i)



No-change + resize controls

Review identity and forensic preservation

Balanced pair anchors each step

3. Shared-weight consistency

Original and approved edit → same detector



Original x



Approved edit x_{edit}

Same-weight detector (encoder + heads + gates + fusion)

Original and edited scores

Classification: ORIGINALS ONLY

Consistency: BOTH scores

$$\mathcal{L}_{inv} = \text{mean}\,|\text{asinh}(z) - \text{asinh}(z_{edit})$$

Train selected encoder stage + heads + gates + fusion

Ablations: BCE → partial fine-tuning → group DRO → age consistency; matched no-change control.

Freeze selection and calibration on validation.
Compare matched test protocols across seeds.